

# Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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**Abstract. Keywords:** Dialect Normalization · Low-Resource NLP · Marathi Dialects · Sequence-to-Sequence Modeling · Synthetic Data Augmentation · mT5 · IndicBART · Text Standardization.

## 1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] have made it possible to develop systems capable of understanding and generating natural language. Open-weight models are used in processing text across various domains involving contextual data as they allow finer control over the model, environment and the data [2, 3]. Even with the given advancements, current systems involving such models tend to cater requests towards standard languages showing undesired output for queries with non-standard dialects [4]. Various surveys conducted [5, 6] have proven that speakers with native dialect have faced several issues within the existing language processing systems such as Automatic Speech Recognition and Machine Translation due to lexical and phonetic variations [7, 8].

Dialect normalization is the task of transducing regional dialectal text into standard written language, where the main difficulty lies in keeping the original meaning intact [8]. It is an important layer for digital linguistic inclusion. As countries like India have 22 officially recognized languages including hundreds of regional dialects spoken by 1.4 billion people, sub-regional varieties have remained unexplored when it comes to computational research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India [9]. In this work, our core motivation stems from analyzing what drives dialectal disparities and why standard models underperform on regional text. We hypothesized that evaluating multi-dialect Marathi text as a single homogenous benchmark masks critical sub-regional nuances. Therefore, we systematically investigate how different regional dialects (Southern Konkan, Northern Konkan, and Varhadi) behave under normalization, proving that evaluation metrics vary significantly across dialects due to distinct lexical, phonetic, and morpho-syntactic properties.

Building pipelines for translating Indic dialects primarily comes down to two core challenges:

1. **Extreme Data Scarcity:** Datasets that map regional dialects to standard forms are technically non-existent; and
2. **Morphological Complexity:** Marathi is a language with rich morphology, where noun cases, postpositional affixes, and verbal tense suffixes exhibit complex variations [10, 11].

Addressing these limitations, this work presents a multi-dialect Marathi text normalization framework combining synthetic data augmentation with sequence-to-sequence transformer fine-tuning. To overcome severe data scarcity while ensuring that synthetic augmentation yields high accuracy of results without introducing noise, we implement an automated multi-tier verification engine (combining script filtering and LLM semantic auditing). Furthermore, we rigorously evaluate the performance of existing pre-trained multilingual sequence-to-sequence models—specifically google/mT5-small (300M) and IndicBART (244M)—to benchmark their capability in capturing complex dialectal shifts.

Our core contributions are:

- **Parallel Corpus Generation:** Based on the transcripts of RESPIN dataset [12], we construct a parallel corpus of three Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi across agricultural and banking domains.
- **Automated Synthetic Augmentation with Verification:** We implement a synthetic data expansion pipeline using Gemma-4 [3] with a verification step that flags and removes corrupted pairs, proving that verified synthetic augmentation consistently provides higher accuracy of results.
- **Cross-Dialect & Multilingual Model Benchmarks:** We provide a comprehensive empirical evaluation of existing multilingual models (mT5-small vs. IndicBART), demonstrating that evaluation results vary significantly across individual dialects (e.g., up to 81.00 BLEU on Varhadi) and establishing benchmark standards for Indic dialect normalization.

## 2 Related Work

### 2.1 Regional Dialect Standardization across Language Families

Early approaches in the context of European languages used Character-Level Statistical Machine Translation (CSMT) and phrase-based models for Swiss German Dialects [13] which were later replaced by pre-trained language models [14, 15]. Similar approaches have been developed for Finnish, Swedish, and Estonian dialects [11, 16, 17]. A study of four language families by Kuparinen et al. [8] suggested that fine-tuned sequence-to-sequence transformers surpassed the rule-based and SMT baselines. Other than European countries, East Asian and African-Asian varieties have also shown similar patterns. Japanese regional dialects have been processed using phrase-level LSTM modules [18], while Mandarin scripts relied on acoustic-phonetic mapping [19]. The Conventional Orthography for Dialectal Arabic (CODA) framework [20] in the landscape of Arabic languages gave guidelines for Modern Standard Arabic (MSA) which led to

multi-city translation benchmarks (MADAR [21], NADI [22]) and morphology-oriented prompt design for LLMs [23, 24].

## 2.2 Indic Dialect Landscape and Marathi NLP

Even though pre-trained models such as IndicBART [9] and mT5 [25] which are trained on multilingual corpora cover major languages, they lack understanding of regional sub-dialects. Models based on a single language like L3Cube-MahaNLP [26] offer pre-trained marathi models for classification, but do not support generative text standardization. INDIC-DIALECT [27] considers Hindi varieties (Bhojpuri, Magahi, Maithili), for evaluation via WFST frameworks for text normalization while preserving the underlying semantics [28]. However, Marathi dialects were primarily dealt for acoustic/ASR classification [29, 30], or zero-shot LLM retrieval [31, 32]. Our work fills this gap by presenting an end-to-end neural fine-tuning pipeline with evaluations and data augmentation strategies.

## 2.3 LLM Distillation and Multi-Tier Verification

LLMs can be used to suffice for data scarcity in various tasks. [33–35]. Knowledge distillation strategies can be employed to train compact and efficient models based on the outputs generated by a larger teacher model. [36–38]. But the use of such techniques for low-resource regional dialects involves significant amount of hallucinations and noise. [38, 39]. We solve this by utilising Gemma-4 with a robust morphologically aware system prompt and a multi-tier verification system that ensures integrity of the data generated.

# 3 Methodology

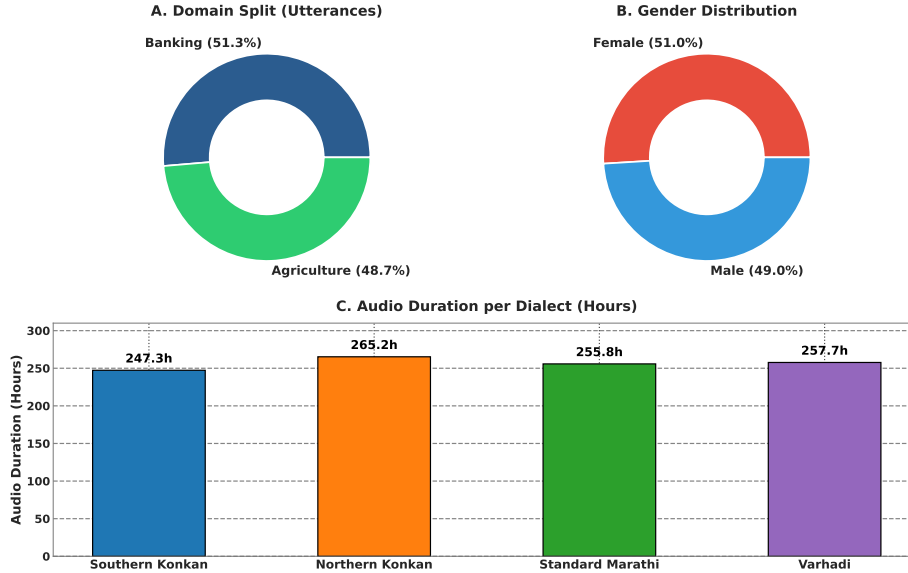
## 3.1 Target Dialects and Linguistic Properties

The following Marathi dialects based on RESPIN corpus were on our radar that exhibit unique lexical and phonetic variations:

1. **Southern Konkani:** Shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms are its key characteristics. Lexical differences include vowel shifts: standard जातो (“he goes”) becomes जातों; standard कुठे (“where”) becomes कुठंय.
2. **Northern Konkani:** Since it is surrounded by neighboring Gujarati and Bhili language families, it tends to exhibit significant consonant variations. Few examples are: standard कशाला (“why”) → Northern Konkani कसाले; standard मुलगा (“boy”) → Northern Konkani पोर.
3. **Varhadi:** Spoken in the regions of Vidarbha accompanied with verb suffix modifications. Standard questioning का? (“why?”) becomes काऊन?; verbal forms exhibit noticable patterns. Varhadi maintains closest structural similarity to Standard Marathi as compared to other dialects since the transformations are more systematic.

**Table 1.** RESPIN-S1.0 Marathi Source Corpus Statistics

| Code                      | Dialect Name               | Utterances (%)        | Duration (h)    | Speakers     | Unique Prompts |
|---------------------------|----------------------------|-----------------------|-----------------|--------------|----------------|
| D1                        | Southern Konkani (Malvani) | 203,651 (25.14%)      | 247.31          | 732          | 5,838          |
| D2                        | Northern Konkani (Ahirani) | 198,560 (24.52%)      | 265.25          | 557          | 5,825          |
| D3                        | Standard Marathi (Puneri)  | 208,850 (25.79%)      | 255.79          | 608          | 6,077          |
| D4                        | Varhadi (Vidarbha)         | 198,873 (24.55%)      | 257.70          | 747          | 5,330          |
| <b>Total All Dialects</b> |                            | <b>809,934 (100%)</b> | <b>1,026.06</b> | <b>2,644</b> | <b>23,069</b>  |

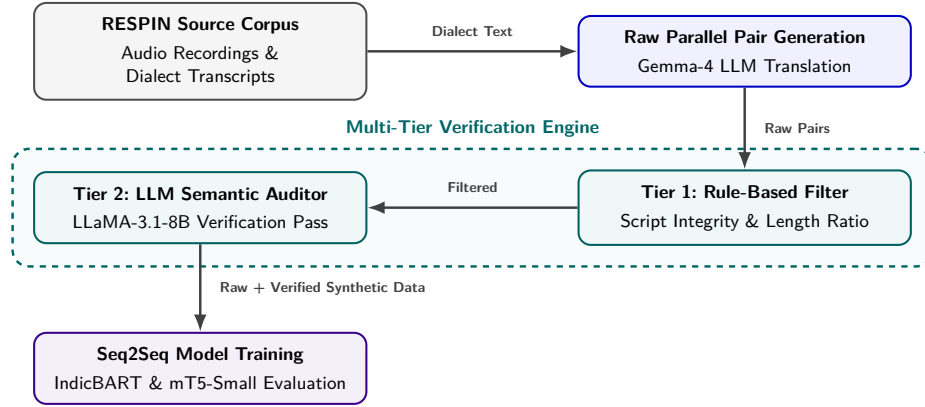
**Fig. 1.** RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution ; (B) gender split ; (C) audio recording duration per dialect variant.

### 3.2 Original Corpus Curation and Foundation in RESPIN

Our primary corpus still remains the domain-specific transcripts of the RESPIN-S1.0 initiative [12], developed by researchers at IISc Bangalore. RESPIN has audio recordings and their transcripts around agriculture and banking domains for different Indian languages and their dialects, it still lacks aligned dialect-to-target language translations which is needed for normalisation.

### 3.3 LLM-Driven Corpus Generation and Verification

We considered several design decisions to ensure generation quality to be maximised: To deal with the unavailability of data we developed a synthetic data generation pipeline considering a few design aspects in order to ensure the maximum quality of data generation:



**Fig. 2.** System architecture of the multi-dialect normalization framework.

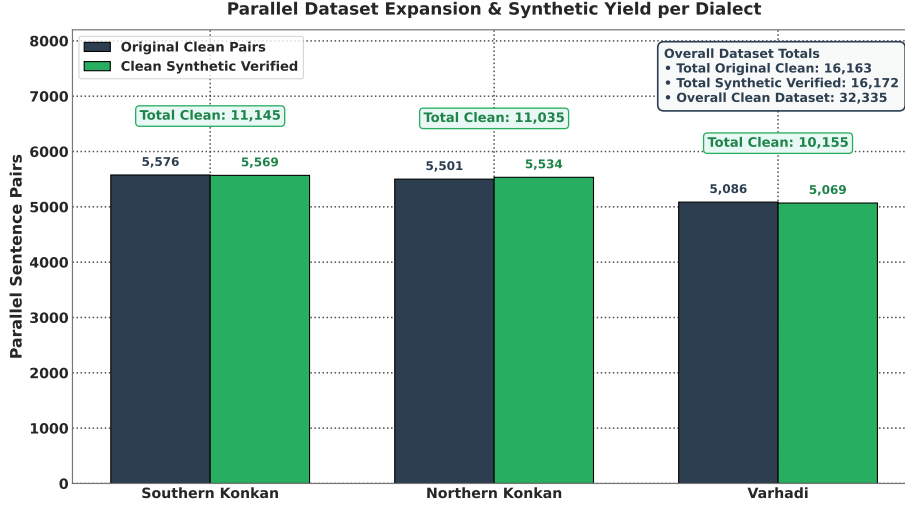
- **1-Prompt-Per-Sentence Architecture:** Instead of batching multiple pairs per API call (which risks contextual loss), each sentence is processed separately.
- **Detailed Prompting Strategy:** System prompt is designed for the output to preserve exact meaning, context regarding the domain, proper nouns, and numerical values from the input with a set of 5-8 few-shot examples per dialect.
- **Temperature Control:** Generation temperature is set to 0.3 with top- $p = 0.9$  for optimal results.

This set was generated using Gemma-4 [3] under the 12B configuration accessed via Ollama, run locally. All generated pairs undergo a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.**
  - *Devanagari Script Integrity:* Both source and target must have at least 90% Devanagari Unicode characters that lie in the U+0900--U+097F range, that automatically get rid of unsolicited characters.
  - *Length Ratio Constraint:*  $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$  to avoid hallucinations.
  - *Non-Identity Constraint:*  $L_{\text{dialect}} \neq L_{\text{standard}}$  to flag identical string copies.
  - *Minimum Token Count:* Both source and target must contain at least 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary round of evaluation was performed using `llama-3.1-8b-instant` (accessed via Groq API). The verification happens on a scale of 1-5 score that considers semantics, hallucinations, multilingual contamination; and complete standardization of dialects. Pairs that score  $< 4.0$  are rejected or routed for further correction.

**Table 2.** Statistics based on generated and verified data

| Dialect Name    | Original Clean Pairs | Raw Synthetic Generated | Clean Synthetic Verified | Corrupted / Rejected Pairs (%) | Total Clean Expanded Pairs |
|-----------------|----------------------|-------------------------|--------------------------|--------------------------------|----------------------------|
| Southern Konkan | 5,576                | 5,838                   | 5,569                    | 269 (4.61%)                    | 11,145                     |
| Northern Konkan | 5,501                | 5,825                   | 5,534                    | 291 (5.00%)                    | 11,035                     |
| Varhadi         | 5,086                | 5,330                   | 5,069                    | 261 (4.90%)                    | 10,155                     |
| <b>Total</b>    | <b>16,163</b>        | <b>16,993</b>           | <b>16,172</b>            | <b>821 (4.83%)</b>             | <b>32,335</b>              |

**Fig. 3.** Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

- Closed-Loop Correction Engine.** Pairs that are flagged by Tier 1 or Tier 2 enter a correction loop: Step 1 generates corrected translations using specialised prompts based on pre-defined reason of identified failure (e.g., “Hindi leakage detected” or “Incomplete normalization”); Step 2 verifies the corrected sentences again, following the complete discard of pairs that fail the second audit pass.

The given pipeline was used to generate parallel pairs from the RESPIN transcripts. The same pipeline was used further for generation of additional synthetic data for expansion. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as mentioned in Table 2. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified parallel sentence pairs across all three dialects.

**Table 3.** Comprehensive Evaluation Matrix on IISc\_RESPIN\_test\_mr Benchmark Set (2,170 Utterances)

| Model Scope    | Training Data          | Dialect / Subset   | Utts | IndicBART (244M) |               | mT5-Small (300M) |               |
|----------------|------------------------|--------------------|------|------------------|---------------|------------------|---------------|
|                |                        |                    |      | BLEU (↑)         | WER (↓)       | BLEU (↑)         | WER (↓)       |
| Single-Dialect | Original               | Southern Konkan    | 559  | <b>57.80</b>     | <b>26.60%</b> | 43.86            | 34.37%        |
|                |                        | Northern Konkan    | 540  | <b>90.13</b>     | <b>6.46%</b>  | 79.46            | 11.95%        |
|                |                        | Varhadi            | 516  | <b>83.59</b>     | <b>10.19%</b> | 74.81            | 14.73%        |
|                | Synthetically Expanded | Southern Konkan    | 559  | 24.06            | 60.32%        | <b>44.36</b>     | <b>32.75%</b> |
|                |                        | Northern Konkan    | 540  | 65.41            | 28.70%        | <b>79.76</b>     | <b>12.61%</b> |
|                |                        | Varhadi            | 516  | 67.97            | 25.43%        | <b>76.90</b>     | <b>14.19%</b> |
| Multi-Dialect  | Original               | Southern Konkan    | 559  | 26.08            | 52.98%        | <b>40.94</b>     | <b>35.34%</b> |
|                |                        | Northern Konkan    | 540  | 47.92            | 32.64%        | <b>77.50</b>     | <b>14.65%</b> |
|                |                        | Standard Benchmark | 555  | 96.12            | 3.12%         | <b>96.65</b>     | <b>1.91%</b>  |
|                |                        | Varhadi            | 516  | 73.04            | 26.96%        | <b>73.47</b>     | <b>16.16%</b> |
|                | Synthetically Expanded | Southern Konkan    | 559  | <b>52.15</b>     | 36.50%        | 43.88            | <b>34.96%</b> |
|                |                        | Northern Konkan    | 540  | 54.35            | 25.65%        | <b>78.16</b>     | <b>13.55%</b> |
|                |                        | Standard Benchmark | 555  | 96.80            | 1.65%         | <b>97.39</b>     | <b>1.37%</b>  |
|                |                        | Varhadi            | 516  | 69.96            | 21.04%        | <b>74.74</b>     | <b>15.57%</b> |

### 3.4 Evaluation Protocol

For stable performance during development, models were validated using 5-Fold Cross-Validation on an 85/15 stratified split of the train set. The final reports were generated on the official held-out test set provided by RESPIN after training the model on the complete dataset, which had around 2,170 utterances across target dialects. This setup is applied across all dialects on the original transcript-based data and their synthetically expanded versions. Results are reported using metrics that are evaluated using SacreBLEU [40] and Word Error Rate (WER) scripts:

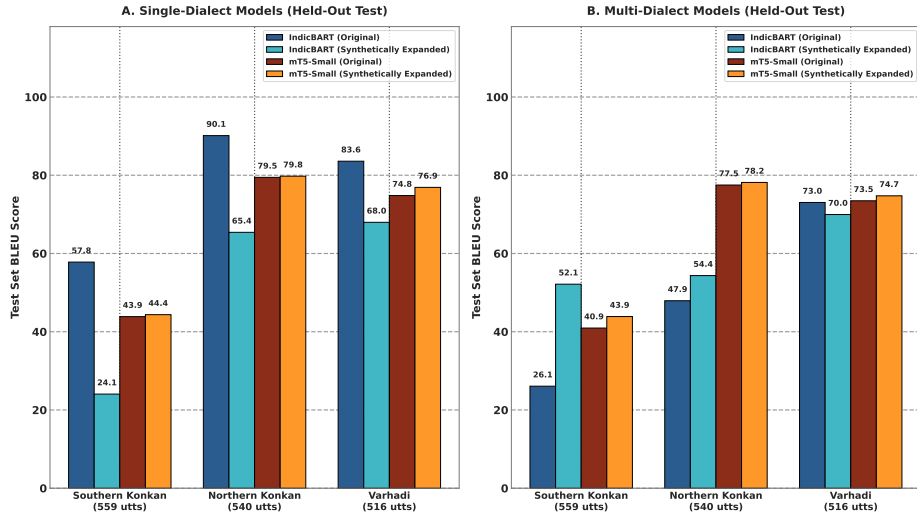
- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bi-grams.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

## 4 Findings

### 4.1 Comprehensive Test Benchmark on IISc\_RESPIN\_test\_mr

Key observations from Table 3 and Figure 4 include:

1. **mT5-Small Performance**: mT5-small upon finetuning on the synthetically expanded dataset achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling**: Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, suggesting that larger data size brings up the efficiency in mBART-50 architectures.



**Fig. 4.** Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

**Table 4.** Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

| Dialect Name          | Total Pairs | Test Set Pairs (IB / mT5) | IndicBART (244M) |        | mT5-Small (300M) |        |
|-----------------------|-------------|---------------------------|------------------|--------|------------------|--------|
|                       |             |                           | BLEU             | chrF++ | BLEU             | chrF++ |
| Southern Konkani (D1) | 5,576       | 837 / 836                 | 48.54            | 78.35  | 46.31            | 74.43  |
| Northern Konkani (D2) | 5,501       | 826 / 825                 | 60.58            | 80.30  | 60.31            | 77.34  |
| Varhadi (D4)          | 5,086       | 763 / 762                 | 76.63            | 90.93  | 81.00            | 91.21  |

#### 4.2 Single-Dialect vs. Multi-Dialect Benchmark Results on Original and Synthetic Datasets

Table 4 presents the performance matrix on original single-dialect benchmarks, and Table 5 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 6 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 7 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

#### 4.3 Impact of Synthetic Data Augmentation on Test Set

Table 8 and Figure 7 show the overall impact on metrics by synthetic data augmentation directly on the held-out test set under multi-dialect training schemes. IndicBART showed improvement for Southern Konkani (+26.07 BLEU) and



**Table 5.** Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

| Evaluated Subset        | Total Pairs   | Test Set (IB / mT5)  | IndicBART (244M) |              | mT5-Small (300M) |              |
|-------------------------|---------------|----------------------|------------------|--------------|------------------|--------------|
|                         |               |                      | BLEU             | chrF++       | BLEU             | chrF++       |
| Southern Konkani (D1)   | 16,163        | 837 / 836            | 26.21            | 53.15        | <b>48.28</b>     | <b>75.04</b> |
| Northern Konkani (D2)   | 16,163        | 826 / 798            | 47.59            | 67.16        | <b>62.35</b>     | <b>78.92</b> |
| Varhadi (D4)            | 16,163        | 763 / 790            | 72.75            | 85.63        | <b>79.57</b>     | <b>90.51</b> |
| <b>Overall Combined</b> | <b>16,163</b> | <b>2,426 / 2,424</b> | <b>48.17</b>     | <b>68.58</b> | <b>63.29</b>     | <b>81.37</b> |

**Table 6.** Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

| Dialect Name          | Total Pairs | Test Set (IB / mT5) | IndicBART (244M) |        | mT5-Small (300M) |              |
|-----------------------|-------------|---------------------|------------------|--------|------------------|--------------|
|                       |             |                     | BLEU             | chrF++ | BLEU             | chrF++       |
| Southern Konkani (D1) | 11,145      | 1,672 / 1,671       | 47.25            | 64.69  | <b>65.10</b>     | <b>82.88</b> |
| Northern Konkani (D2) | 11,035      | 1,656 / 1,655       | 40.51            | 62.21  | <b>62.07</b>     | <b>79.29</b> |
| Varhadi (D4)          | 10,155      | 1,524 / 1,523       | 73.62            | 86.75  | <b>78.89</b>     | <b>90.59</b> |

Northern Konkani (+6.43 BLEU), but its Varhadi score dropped by 3.08 BLEU. mT5-small showed consistent improvements: +2.94 BLEU for Southern Konkani, +0.66 BLEU for Northern Konkani, and +1.27 BLEU for Varhadi.

#### 4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

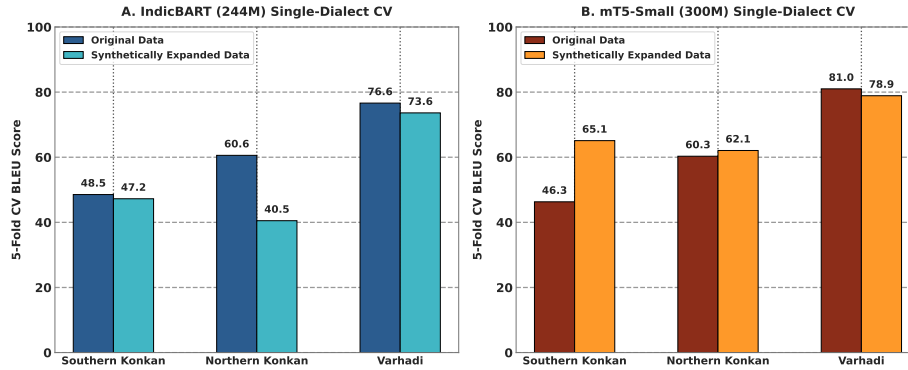
Performance of our verification and filtering engine is measured by conducting a comparative analysis on models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite). The raw unverified dataset contains 19,914 parallel pairs, that included 3,751 rejected pairs that contained garbage translations. Table 9 presents the quantitative comparison on the benchmark suite.

Key findings and insights from this evaluation include:

1. **Quality Collapse on Unverified Data:** Training IndicBART on unverified outputs causes a significant drop of 14.03 BLEU points, collapsing from 57.12 to 43.09 due to compounded error propagation from uncorrected dialect residue and wrong syntax with other errors.
2. **mT5-Small Resilience:** While google/mT5-small showed resilience due to its 250,000-token multilingual capacity, training on unverified data still suffers a 8.30 BLEU point drop, falling from 69.51 to 61.21 and a 4.40 chrF++ drop.
3. **Essential Role of Multi-Tier Auditing:** Removal of flawed ones and correcting the possible set of pairs is directly responsible for better performance that eliminated subsequent errors.

**Table 7.** Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

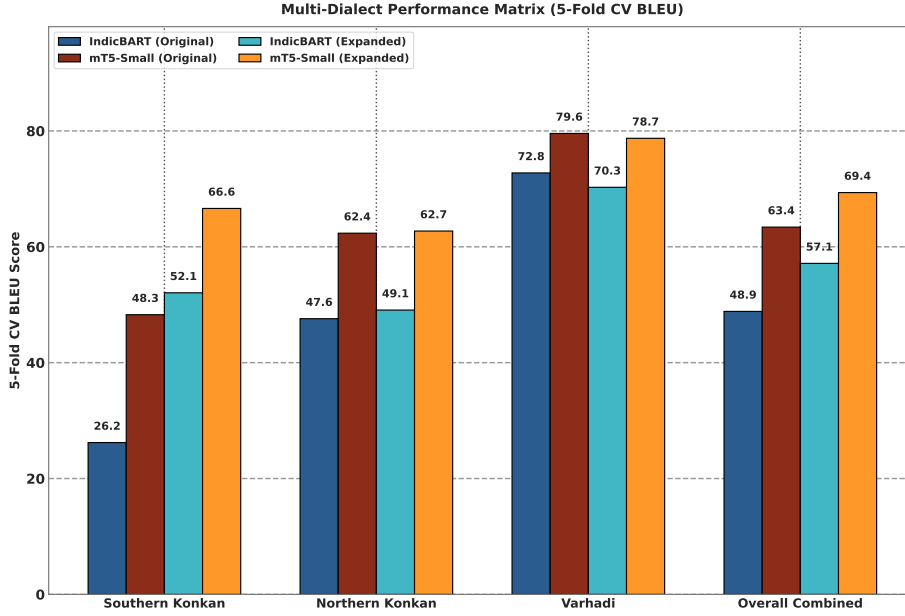
| Evaluated Subset        | Total Pairs   | Test Set (IB / mT5)  | IndicBART (244M) : mT5-Small (300M) |              |              |              |
|-------------------------|---------------|----------------------|-------------------------------------|--------------|--------------|--------------|
|                         |               |                      | BLEU                                | chrF++       | BLEU         | chrF++       |
| Southern Konkani (D1)   | 32,335        | 1,672 / 1,710        | 52.06                               | 72.15        | <b>66.62</b> | <b>83.57</b> |
| Northern Konkani (D2)   | 32,335        | 1,656 / 1,627        | 49.08                               | 69.90        | <b>62.72</b> | <b>79.63</b> |
| Varhadi (D4)            | 32,335        | 1,524 / 1,513        | 70.27                               | 84.27        | <b>78.73</b> | <b>90.46</b> |
| <b>Overall Combined</b> | <b>32,335</b> | <b>4,852 / 4,850</b> | <b>57.12</b>                        | <b>75.49</b> | <b>69.51</b> | <b>84.58</b> |

**Fig. 5.** Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

#### 4.5 Model Architecture Analysis

Table 10 provides detailing on the architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage:** mT5’s 250K vocabulary provides  $3.9\times$  more subword units than IndicBART’s 64K, enabling finer-grained tokenization of morphologically complex Marathi dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.
2. **Positional Encoding:** mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead:** IndicBART requires explicit language tokens (`<2mr>`) that add conditioning complexity, whereas mT5’s pure text-to-text format enables direct mapping without task-specific architectural constraints.



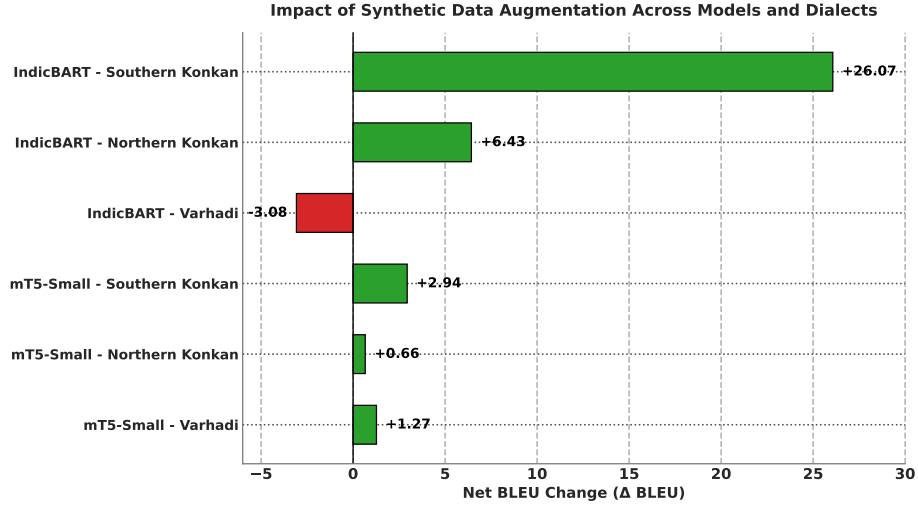
**Fig. 6.** Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact ( $\Delta$  BLEU across sub-dialects).

**Table 8.** Effect of Data Augmentation on Held-Out Test BLEU

| Evaluated Subset     | IndicBART BLEU |          |               | mT5-Small BLEU |              |              |
|----------------------|----------------|----------|---------------|----------------|--------------|--------------|
|                      | Original       | Expanded | $\Delta$      | Original       | Expanded     | $\Delta$     |
| Southern Konkan (D1) | 26.08          | 52.15    | <b>+26.07</b> | 40.94          | <b>43.88</b> | <b>+2.94</b> |
| Northern Konkan (D2) | 47.92          | 54.35    | <b>+6.43</b>  | 77.50          | <b>78.16</b> | <b>+0.66</b> |
| Varhadi (D4)         | <b>73.04</b>   | 69.96    | -3.08         | 73.47          | <b>74.74</b> | <b>+1.27</b> |

**Table 9.** Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

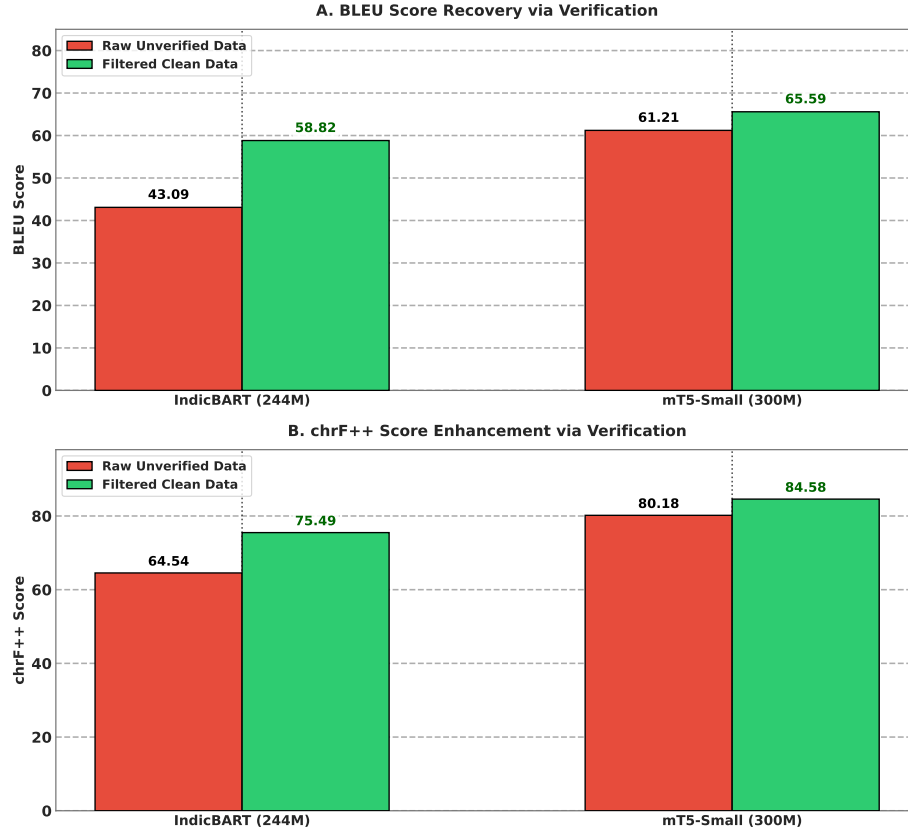
| Model            | Pipeline State      | Total Pairs | BLEU         | chrF++       | $\Delta$ BLEU | $\Delta$ chrF++ |
|------------------|---------------------|-------------|--------------|--------------|---------------|-----------------|
| IndicBART (244M) | Raw Unverified Data | 19,914      | 43.09        | 64.54        | -14.03        | -10.95          |
|                  | Filtered Clean Data | 32,335      | <b>57.12</b> | <b>75.49</b> | <b>+14.03</b> | <b>+10.95</b>   |
| mT5-Small (300M) | Raw Unverified Data | 19,914      | 61.21        | 80.18        | -8.30         | -4.40           |
|                  | Filtered Clean Data | 32,335      | <b>69.51</b> | <b>84.58</b> | <b>+8.30</b>  | <b>+4.40</b>    |



**Fig. 7.** Impact of LLM synthetic data expansion ( $\Delta$  BLEU score comparing synthetically expanded vs. original benchmarks on the held-out test set).

**Table 10.** Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

| Property            | IndicBART (244M)                     | mT5-small (300M)               |
|---------------------|--------------------------------------|--------------------------------|
| Base Architecture   | mBART-50 (BART Enc-Dec)              | T5 Encoder-Decoder             |
| Position Encoding   | Absolute sinusoidal                  | Relative positional bias       |
| Vocabulary Size     | 64,000 SentencePiece                 | 250,000 SentencePiece (mC4)    |
| Pre-training Data   | 11 Indic + English                   | 101 languages (mC4)            |
| Script Approach     | Devanagari-unified                   | Native multi-script            |
| Target Conditioning | Required $\langle 2mr \rangle$ token | Raw text-to-text (no prefix)   |
| Learning Rate       | $5 \times 10^{-5}$ (linear warmup)   | $5 \times 10^{-4}$ (AdaFactor) |



**Fig.8.** Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

## 5 Conclusion

This study presents a comprehensive, reproducible framework for multi-dialect Marathi text normalization, addressing severe data scarcity through automated synthetic data expansion and establishing rigorous neural sequence-to-sequence benchmarks.

Our LLM-driven synthetic data augmentation pipeline, powered by Gemma-4 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing), successfully doubled the clean parallel corpus through synthetic expansion while maintaining strict semantic fidelity—a scalable paradigm for other low-resource dialect pairs. Fine-tuning sequence-to-sequence transformers on the expanded benchmark established state-of-the-art performance: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the synthetically expanded multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. Individual dialect performance reached 81.00 BLEU and 91.21 chrF++ for Varhadi, demonstrating that compact pre-trained sequence-to-sequence transformers can achieve high-accuracy dialect normalization.

### 5.1 Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on LLM APIs (Gemma-4 via local/hosted endpoints), introducing API cost and latency considerations.
2. While we benchmarked two encoder-decoder architectures (mT5-small and IndicBART), decoder-only LLMs (e.g., LLaMA-3, Gemma-4) have not been evaluated for zero-shot or fine-tuned text normalization.
3. Evaluation is currently focused on text normalization metrics (BLEU, chrF++, WER); large-scale downstream task evaluation across diverse domain applications remains for future work.

### 5.2 Data and Code Availability

The complete parallel corpus of 32,335 verified dialect–Standard Marathi sentence pairs, along with all training code, evaluation scripts, and pre-trained model checkpoints, will be publicly released upon publication. The dataset will be available under an open license to support reproducible research in Indic dialect normalization. Our synthetic data generation pipeline, multi-tier verification engine, and fine-tuning configurations are documented in the accompanying code repository to enable replication and extension to additional Marathi sub-dialects and other low-resource Indic language varieties.

### 5.3 Future Directions

Several promising directions emerge from this work:

1. **Edge Deployment via Quantization:** Model quantization (INT8/INT4) and knowledge distillation to sub-100M parameter models for on-device deployment in rural Maharashtra.
2. **Expansion to Additional Dialects:** Extending the pipeline to Deshi/Puneri, Kolhapuri, Marathwada, and other Indic language sub-dialects.
3. **Decoder-Only LLM Benchmarking:** Evaluating instruction-tuned decoder-only models for zero-shot and few-shot dialect normalization tasks.

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